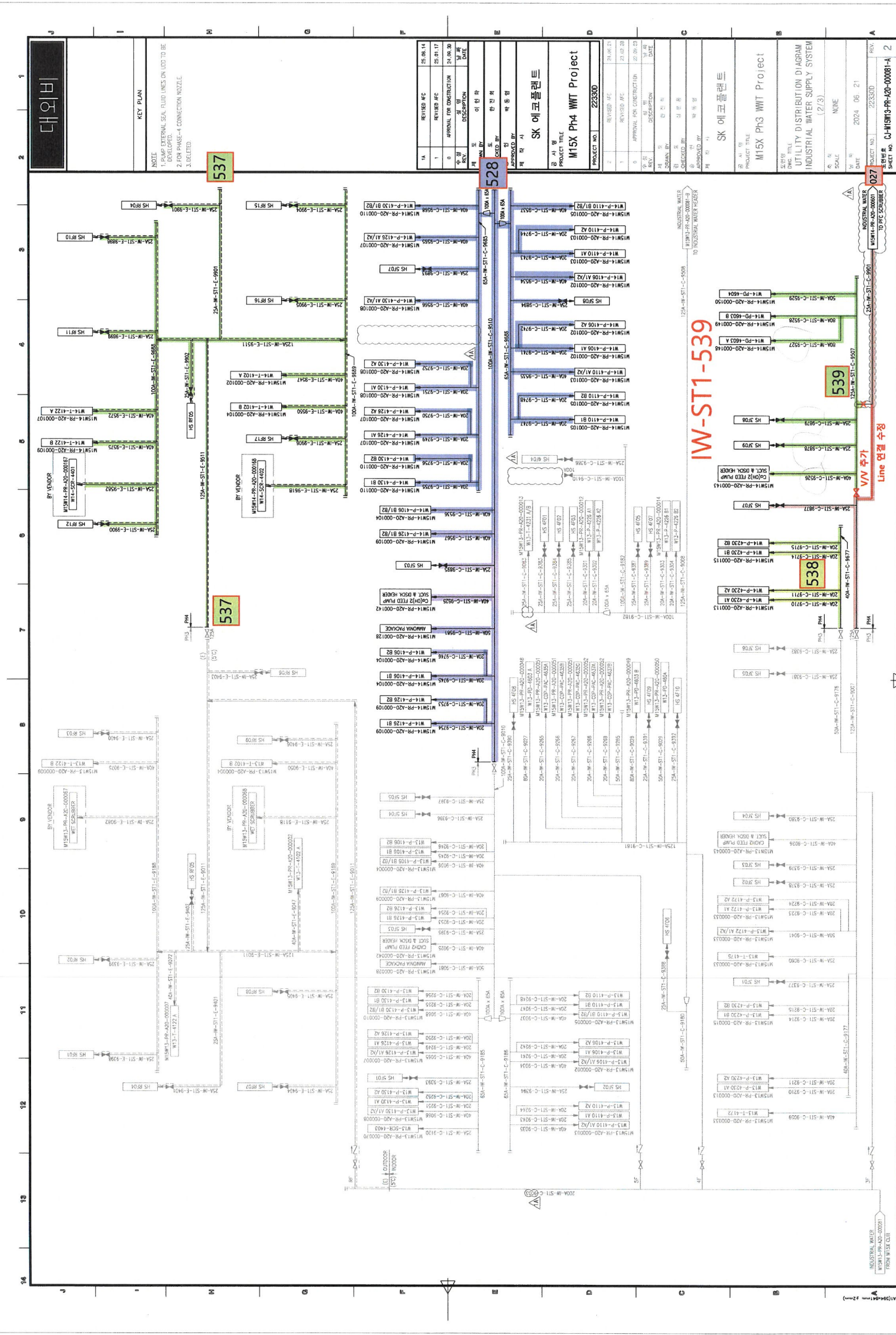
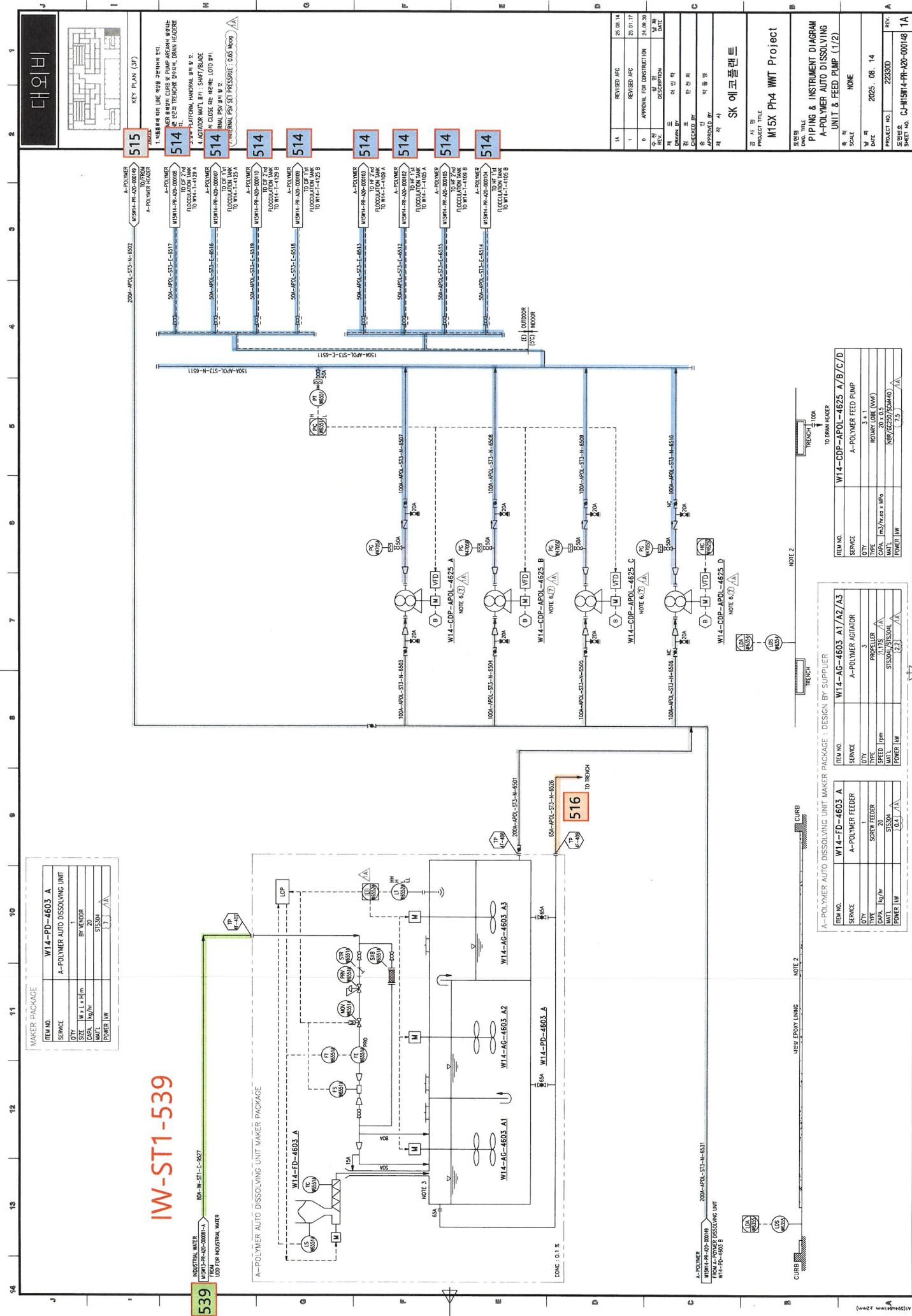






1A	REVISED A/C	25.08.14
1	REVISED A/C	25.01.17
	APPROVAL FOR CONSTRUCTION	24.08.30
주최 REV.	설명 DESCRIPTION	날짜 DATE
제출 SUBMITTED BY	도면 DRAWING	
검토 CHECKED BY	이런리 이런리	
승인 APPROVED BY	박동영 박동영	
제 작 사	SK 에코플랜트	
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도면 명 DRAWING TITLE	PIPING & INSTRUMENT DIAGRAM A-POLYMER AUTO DISSOLVING UNIT & FEED PUMP (1/2)	
스 케 일 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PRODUCT NO.	223300	
도면번호 SHEET NO.	C-15M14-PR-A20-000-08	1A

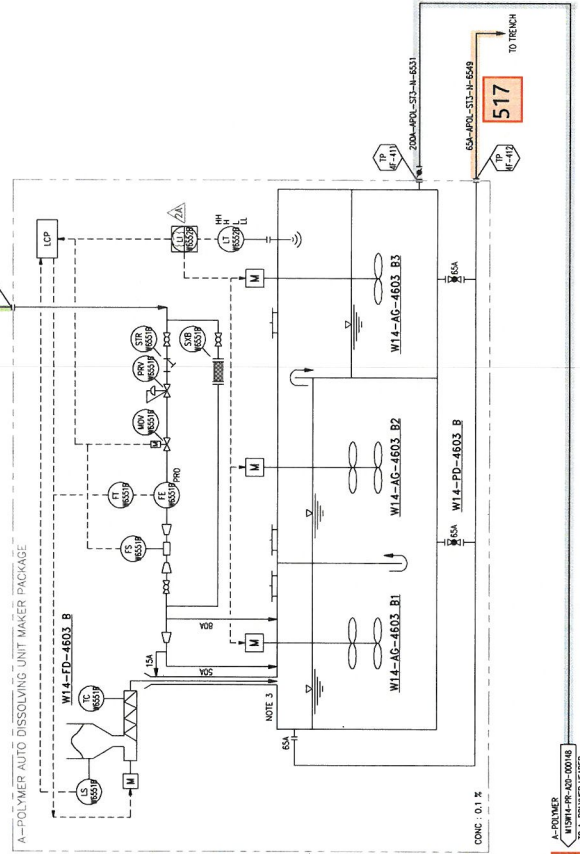
ITEM NO.	W14-COP-APOL-4625 A/B/C/D
SERVICE	A-POLYMER FEED PUMP
QTY	3 ± 1
TYPE	ROTARY LOBE (NMF)
CAP.	20 ± 0.5
MAT'L	NBR/62/25/SUL440
POWER HW	(7.5) 1/A

A - POLYMER AUTO DISSOLVING UNIT MAKER PACKAGE : DESIGN BY			
ITEM NO.	W14-FD-4603 A	ITEM NO.	W14-AQ-4603 A1/A2/A3
SERVICE	A-POLYMER FEEDER	SERVICE	A-POLYMER AGITATOR
QTY	1	QTY	3
TYPE	SOFT FEEDER	TYPE	PROPELLER
GPM	70	SPEED	1/15 A
SYSTEM	SOFT	SYSTEM	5555
POWER KW	0.4 A	POWER KW	12.1 A

IW-ST1-539

MAKER PACKAGE	
ITEM NO.	W14-PD-4603 B
SERVICE	A-POLYMER AUTO DISSOLVING UNIT
QTY	1
SIZE	W x L x H (mm)
CAPA	kg/hr
MATL	SS304
POWER (kW)	1.7 / 2A

539 INDUSTRIAL WATER
W15W14-PR-A20-000149
FROM
LOD FOR INDUSTRIAL WATER



515 A-POLYMER
W15W14-PR-A20-000149
TO A-POLYMER FEEDER

517 65A-APOL-ST1-N-6531
TO TRENCH

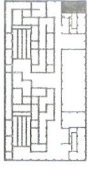
A-POLYMER AUTO DISSOLVING UNIT MAKER PACKAGE : DESIGN BY SUPPLIER	
ITEM NO.	W14-AG-4603 B1/B2/B3
SERVICE	A-POLYMER ACTUATOR
QTY	3
SIZE	W x L x H (mm)
CAPA	kg/hr
MATL	SS304/SS304L
POWER (kW)	0.4 / 2A

ITEM NO.	W14-CDP-APOL-4626 A/B/C/D
SERVICE	A-POLYMER FEED PUMP
QTY	3 + 1
SIZE	W x L x H (mm)
CAPA	m ³ /hr ca x MPa
MATL	NER/CS250/SS304
POWER (kW)	3.7 / 2A

NOTE 2



대요비



KEY PLAN (3/7)

- NOTE
1. 내벽용 배관 라인 설치를 고려하여 설치.
 2. POLYMER 배관 라인 CUBB & PUMP AREA에 설치.
 3. CUBB & PUMP AREA에 설치.
 4. 2W PLATFORM, HANDRAIL 2W B 2.
 5. ACTUATOR MATL B71 : SHFT/BLADE
 6. 2W & CLOSE RIG. HEAVY LOD 2W.
 7. INTERNAL PSY SET PRESSURE 0.05 MPa
 8. INTERNAL PSY SET 2W B 2.

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

443

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 A

A-POLYMER

W15W14-PR-A20-000149

FLOCCULATION TANK

TO W14-1-194 B

200A-APOL-ST1-N-6502

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

A-POLYMER FEEDER

515

A-POLYMER

W15W14-PR-A20-000149

SK 에코플랜트

공 시 명

PROJECT TITLE

MISX Ph4 WWT Project

도면명

PIPELINE & INSTRUMENT DIAGRAM

A-POLYMER AUTO DISSOLVING

UNIT & FEED PUMP (2/2)

SCALE

NONE

날 월

2025. 08. 14

DATE

REV.

222300

PROJECT NO.

222300

REV.

222300

SHEET NO.

CJ-MISX4-PR-A20-000149

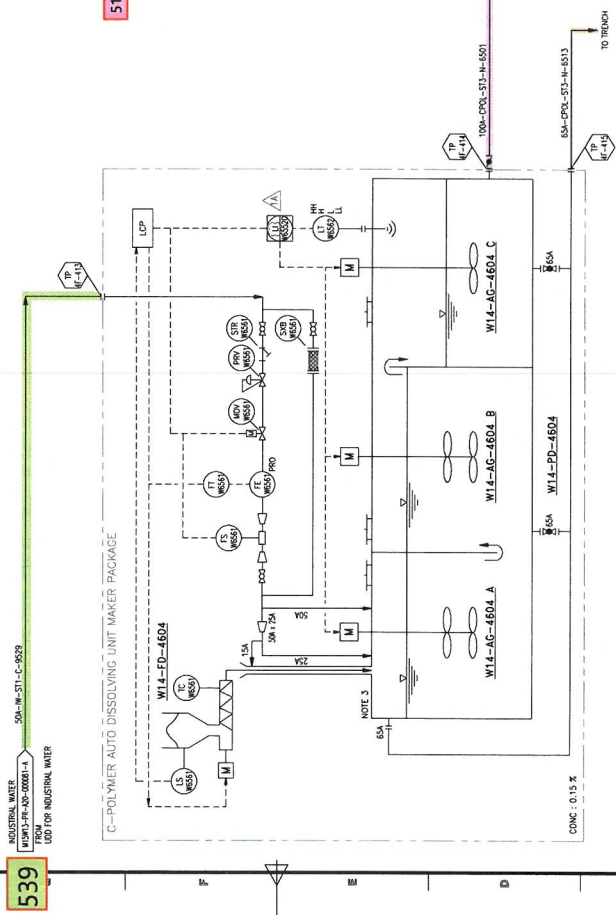
2A

2A

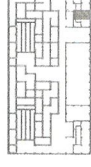
ITEM NO.	W14-PD-4604
SERVICE	C-POLYMER AUTO DISSOLVING UNIT
QTY	1
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304
POWER kW	2.65 / A

MAKER PACKAGE

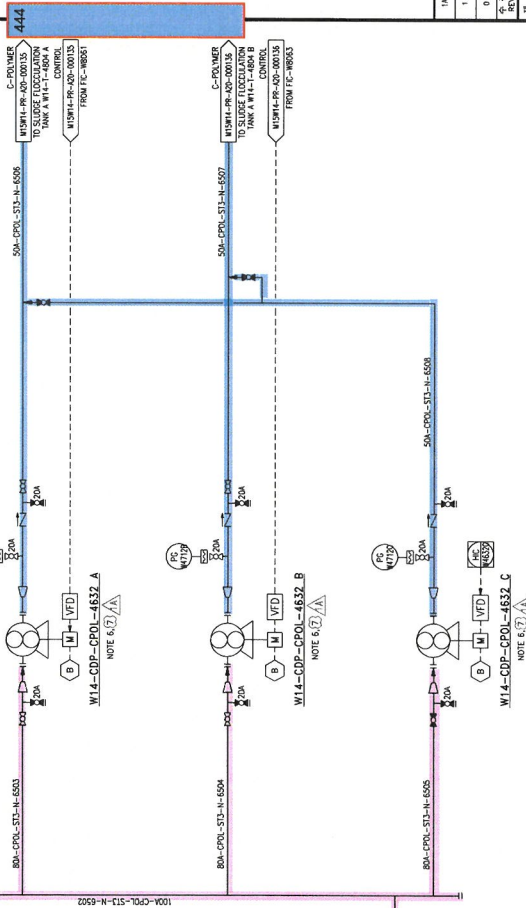
IW-ST1-539



대외비



- NOTE
1. 배관용 재료 LINE 材質을 201 stainless steel 사용한다.
 2. POLYMER 용액의 CURE 및 PUMP AREA에 201 stainless steel 사용한다. DRAIN HEADER 201 stainless steel 사용한다.
 3. 3W PLATFORM, HANDRAIL 201 stainless steel 사용한다.
 4. ACTUATOR MATL BPI - SHAF/BLADE
 5. 201 stainless steel 사용한다. 201 stainless steel 사용한다.
 6. INTERNAL PSV SET PRESSURE : 0.65 MPa
 7. INTERNAL PSV SET PRESSURE : 0.65 MPa



ITEM NO.	W14-FD-4604
SERVICE	C-POLYMER FEEDER
QTY	1
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304
POWER kW	0.4 / A

MAKER PACKAGE

ITEM NO.	W14-AG-4604 A/B/C
SERVICE	C-POLYMER AGITATOR
QTY	3
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304/SS304
POWER kW	0.75 / A

MAKER PACKAGE

ITEM NO.	W14-CDP-CPOL-4632 A/B/C
SERVICE	C-POLYMER FED PUMP
QTY	2 + 1
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304/SS304
POWER kW	2.2 / A

MAKER PACKAGE

ITEM NO.	W14-PR-A20-000150 R1
SERVICE	C-POLYMER AUTO DISSOLVING UNIT & FEED PUMP
QTY	1
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304/SS304
POWER kW	2.2 / A

MAKER PACKAGE

ITEM NO.	W14-PR-A20-000150 R1
SERVICE	C-POLYMER AUTO DISSOLVING UNIT & FEED PUMP
QTY	1
SIZE	W x L x H mm
CAPA	kg/hr
MATL	SS304/SS304
POWER kW	2.2 / A

MAKER PACKAGE



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	125A	47.4M	PIPE, STS 304TP, ERW, BE - SCH 10S
NP1	20A	1EA	NIPPLE, STS 304TP, ERW, 10E - SCH 40S
STU	125AxB0A	1EA	STUB-IN BRANCH
STU	125AxB50A	1EA	STUB-IN BRANCH
HF	40A	1EA	COUPLING HALF, STS304, SW - SCH 10S
HF	25A	4EA	COUPLING HALF, STS304, SW - SCH 10S
HF	20A	1EA	COUPLING HALF, STS304, SW - SCH 10S
PLG	20A	1EA	PLUS-ROUND HEAD, STS304, SCF'D
F	125A	3EA	FLG SO, KS 10K, RF, STS304
FB	125A	1EA	FLG BLIND, KS 10K, RF, STS304
G	125A	3EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM THK
B	125A	3SET	MACHINE BOLT & NUTS, STS 304 / W WASHERS
VB1	20A	1EA	BALL VALVE, KS 10K, SCF'D, BODY-STIS304 / TRIM-STIS304 / SEAT-PTFE, FB, LEVER
IN	25(1X125A	47.5M	관 25(1X125A)
E45	125A	2EA	ELBOW 45 DEG LR, STS316, ERW, BE - SCH 10S
VG	125A	1EA	GATE VALVE, KS 10K, RF, FLG'D, BODY-SSC13 / TRIM-STIS304, BB / OS&Y, HANDWHEEL
LOTO	125A	1EA	LOTO

CLIENT



CONTRACTOR  SK ecoplant

PROJECT NAME

M15X Ph4 Project

100601141110011111

JOB NO. 223300

SH'T NO.	RFV
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01 / 01	1
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[illegible]

INSULATION	CODE THICK.	C	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C.)	AREA	W14
		25	DESIGN	13.5	60	UNIT	3FA
			OPERATING	10	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-030008-A	
						REF. P & ID	25.03.31
						CJ-M15W13-PR-A20-000081-A	REVIS. DATE
							25.01.24
							APPROVED FOR CONSTRUCTION
							REVIS. DATE
							25.01.24
							APPROVED FOR CONSTRUCTION
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							REVIS. DATE
							25.01.24
							APPROVED FOR CONSTRUCTION

경고 : 본 도면은 SK HYUNIX의 기업비밀 자료로서, 어떠한 경우에도 무단의 사용, 복제, 제3자에게 제공, 배포, 유통을 하여서는 안 됩니다. 무단의 사용, 복제, 제3자에게 제공, 배포, 유통을 함을 금지합니다. 이 도면의 무단 사용에 대하여 민/형사상 책임을 지게 됩니다.

[illegible]

CLIENT



2

CONTRACTOR **SK** ecoplant

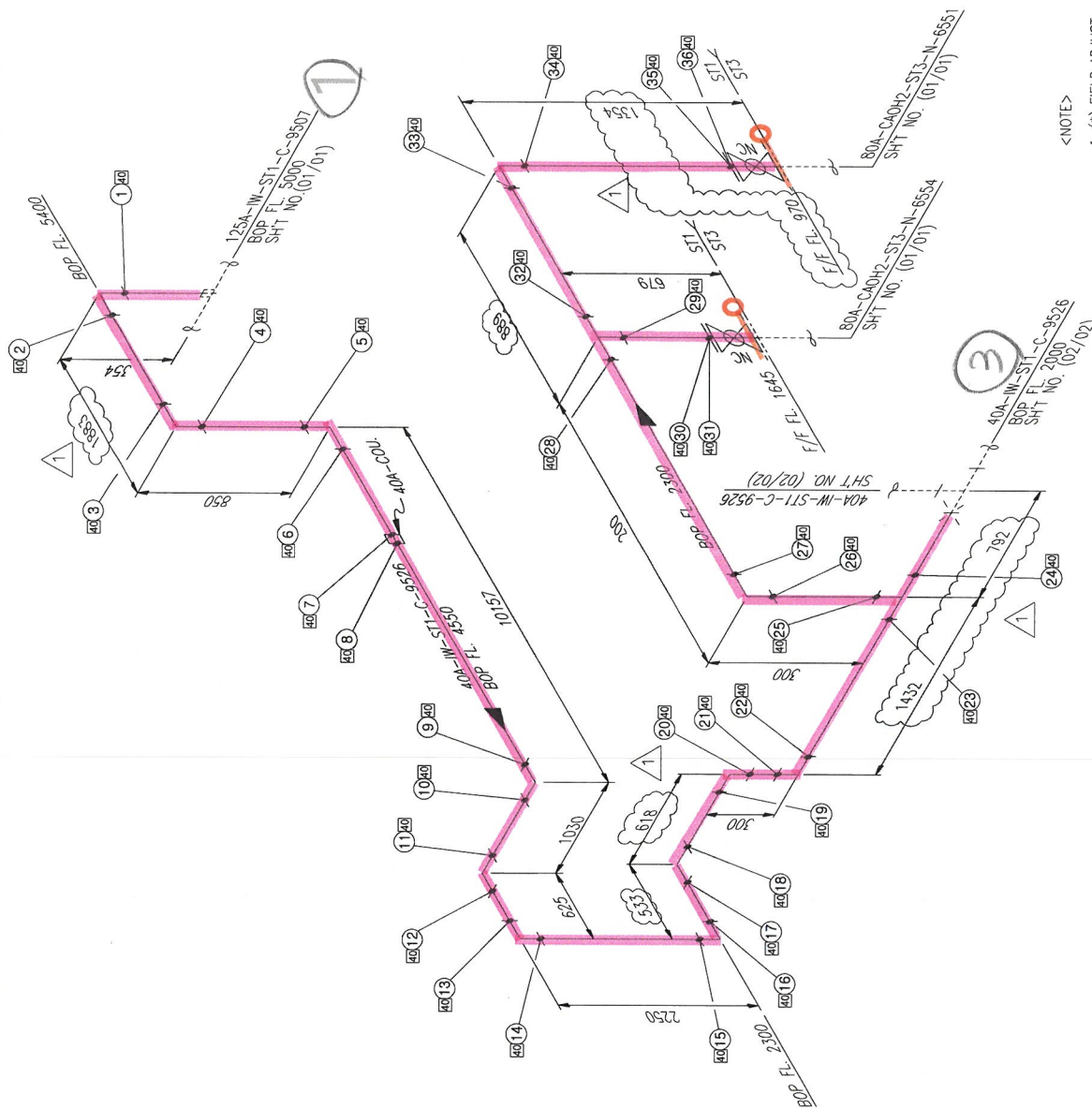
PROJECT NAME
M15X Ph4 Project

JOB NO. _____

22330D

SH'T NO. | REV.

01 / 02	1
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NOTES

1. (*) FIELD ADJUST

PIPE: P/P1	90° ELBOW: Q90	45° ELBOW: F45	TEE: T	REDUCE: TR	STUB-IN: SU	REINFORCING PAD: RP	CONCRETE: RC	FEEDER: FE	CONSUMABLE: RCS/RCS1	WELPLE: WP/NP/NP2	CAP: CAP/CAP1	PLUG: PLG	FULL COUPLING: FC	HAIF COUPLING: HF/H1
STUB-IN: SU	SOCKET/HOPE: SO	SQUAD/HOPE: SQ	FLANGE: F/F1	GASKET: G/G1/G2	BOLT & NUT: B/N	BLIND FLANGE: B/FB1	CATE VAL: VC/MG/MG2	BALL VAL: VB/MG/MG1/MG10/MG20	BUTTERFLY VAL: VB/MG/MG1/MG10/MG20	STEM TRAP: ST	SPRAY NOZZLE: SN	PRESSURE GAUGE: PG	SHIELD: SH	EMERGENCY SHUTTER: SSH
GLOBE VAL: MV/MV1	DWAPDOWN VAL: DV/DV0	DWAPDOWN VAL: DV/DV0	DWAPDOWN VAL: DV/DV0	QUICK COUPLING: SC	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1	FLEXIBLE JOINT: FJ/FJ1
STRAINER: STR	STRAINER: STR	STRAINER: STR	STRAINER: STR	ANCHOR BALL: STOP	BIRD SCREEN: BS	ISOLATION KIT: IS	UNLOADING BOX: UB	AIR TRAP: AT	LOO: LOO	SHIRT GASK: SG	STATION TRAP: ST	SPRAY NOZZLE: SN	PRESSURE GAUGE: PG	SHIELD: SH

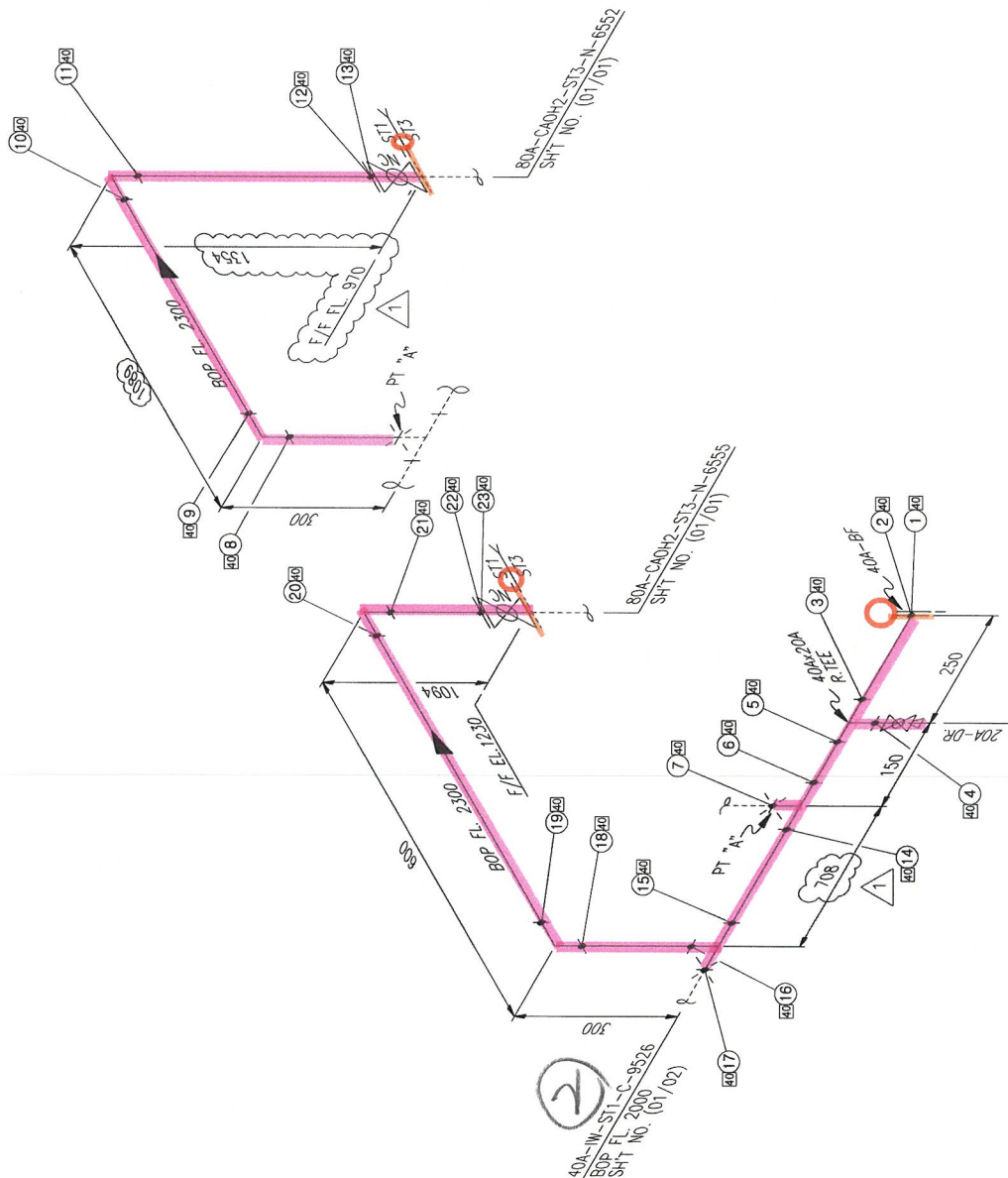
[illegible]

검고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유무의의 폐쇄 사용으로 피하기 발생할 경우 유무의에게 민/형사상 책임/과징금/손상/영양을 요구할 수 있다.

[illegible]

CONTRACTOR  **SK ecoplant**

COMMODITY	IW
LINE NO.	40A-IW-ST1-C-95288



<NOTE>

1. (*) FIELD ADJUST

[illegible][illegible]

경고 : 본 도면은 SK HYUNIX의 기업비밀 자료로서, 어떠한 경우에도 무단 복제, 퍼포, 제3자에게 양도, 대여, 대여, 회신, 또는 도자의 무단 사용을 함을 금하며, 회사에 이를 알리면 상응하는 법적 조치를 취할 권리를 보유하며, 본 도면에 기재된 내용과 관련하여 어떠한 권리도 주장할 수 없습니다.

[illegible]

CLIENT

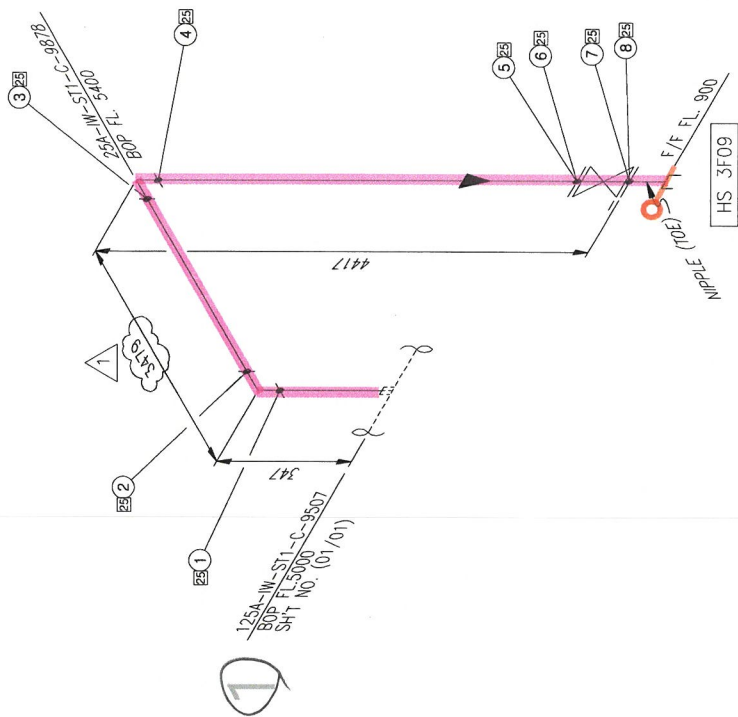


4

CONTRACTOR **SK ecoplant**

PROJECT NAME
M15X Ph4 Project

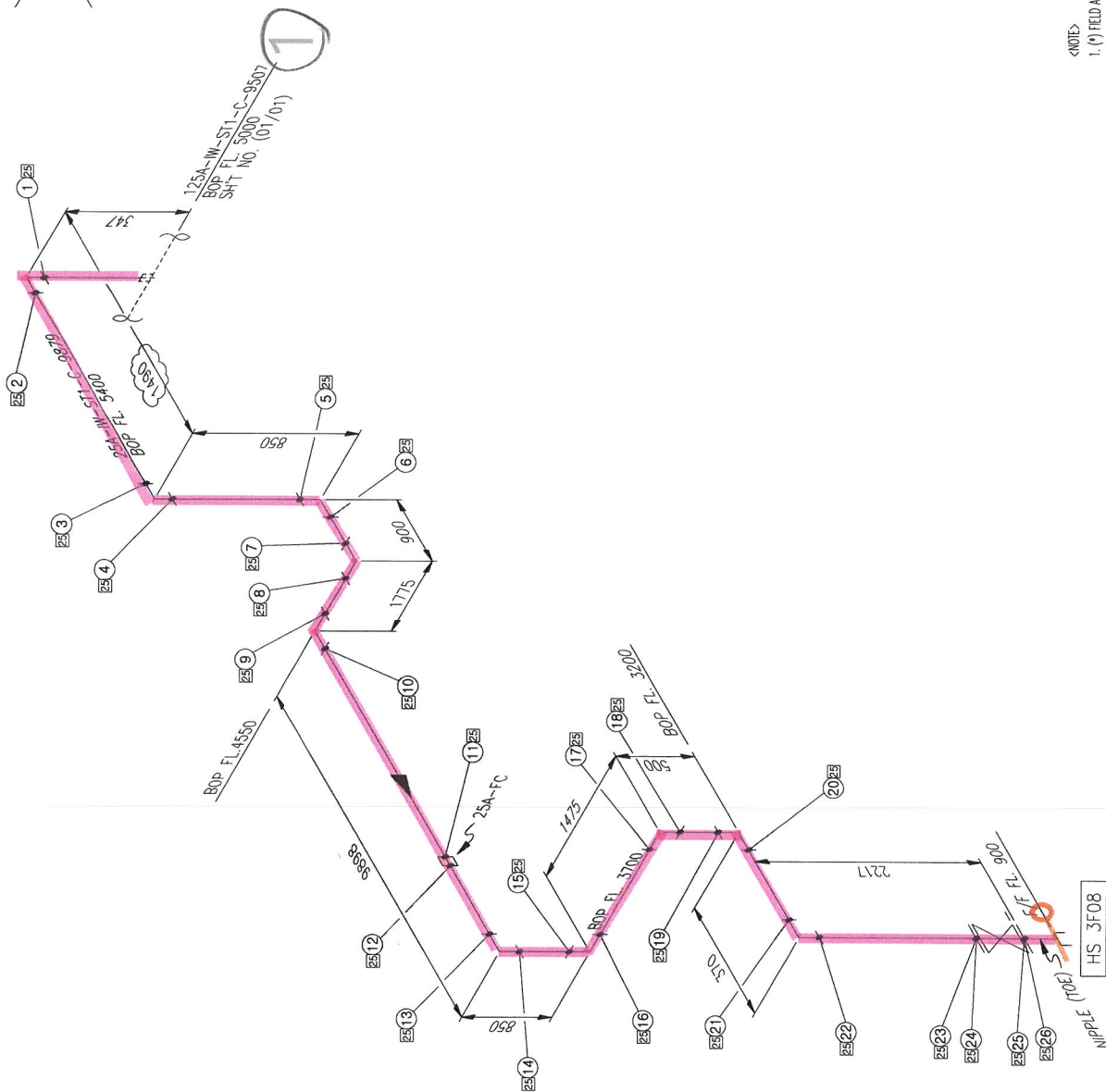
COMMODITY	IW	JOB NO.	22330D
LINE NO.	25A-IW-ST1-C-9878	SH'T NO.	REV.
		01 / 01	1



<NOTE>

* PIPE - P/PI	* 90° ELBOW - E/5	* TEE - T	* ROTATE - TR	* REMOVING PAD - RP	* CONNECT - RC	* COCCED - RE	* CONSUME - RCS/ACS	* FCS SOURCE - RCS/ACS	* WPELE - W/P/W/IN/2	* CAP - CAP/CAP1	* FULL COUPLING - FC	* HALF COUPLING - HF/HF1
* SUB-IN - SU	* SOCKET/PLUG/PC - S	* FLANGE - F/F1	* GASKET - G/G1/G2	* BOLT & NUT - B/N	* BEND LANCE - B/LN	* GATE VA. - G/VG/VA/2	* BALL VA. - B/VG/VA/2	* BALL VA. - B/VG/VA/2	* BUTTERFLY VA. - B/VA	* CHECK VA. - C/VG/VA/2/VA/MS/VA/2	* STEAM TRAP - ST	* SPRAY NOZZLE - SPN
* VALVE - V/V1/V2	* FLEXIBLE JOINT/PIPER - F/J	* FLEXIBLE JOINT/PIPER - F/J	* UNLOADING BKT - U	* ISOLATION KIT - I	* T/O/D - T/O/D	* SIGHT GLASS - SG	* PRESSURE GAUGE - PG	* ANCHOR PLATE - AP	* BRD SCREEN - BS	* STANCHION (PIPE) - STP	* STANCHION (PIPE) - STP	* STANCHION (PIPE) - STP

정규 : 편도면은 SK HYUN의 기업비밀 자료로서, 어떠한 경우에도 전자에게 제3자에게 공개, 복제, 배포할 수 없습니다. 이점 유의하시기 바랍니다.



<NOTE>
1. (*) FIELD ADJUST

HS 3F08

[illegible][illegible]

정고 : 본 도연은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 무단 사용, 복제, 제3자에게 공개, 누출 또는 배포할 수 없으며, 회사나 그 밖의 자에게도 제공하지 않아야 함을 청탁함

CLIENT	 5
CONTRACTOR	
COMMODITY	IW
PROJECT NAME	M15X Ph4 Project
JOB NO.	223300D
LINE NO.	25H-IW-ST1-C-9879
SHT NO. / REV.	01 / 01 1



<NOTE>
1. (*) FIELD ADJUST

[illegible]

CLIENT



9

INSULATION	CODE THICK.	C	CONDITION	PRESS. (kg/cm ²)	TEMP. (°C.)	AREA	Wt14
		25	DESIGN	BY VENDOR	60	UNIT	3FA
			OPERATING	BY VENDOR	32	REF. PLAN DWG.	
						CJ-M15W14-PI-A30-030009-C	
						REV. P & ID	
						CJ-M15W13-PR-A20-000081-A	

CONTRACTOR	PROJECT NAME
SK ecoplast	M15X P4 Project

COMMODITY	JOB NO.
IW	223300

LINE NO.	SHT NO.	REV.
80A-IW-STI-C-9527	01 / 01	1

BY	CHK	APPROV'D
한보수	한보수	박동영

REVISION DATE	DESCRIPTION
25.01.24	APPROVED FOR CONSTRUCTION

DATE	DESCRIPTION
25.01.31	REVISED AS MARKED

정고 : 판 도면은 SK HYUN의 기업비밀 자료로서, 어떠한 경우에도 무단 전·사 용, 복제, 제3자에게 제공, 누출 또는 폐기할 수 없으며, 회사 내의 모든 직원 및 임원, 사원, 계약자, 협력업체, 고객, 대리점, 판매처, 기타 이해관계자 등에게도 이 점에 대하여 교육시켜야 하며, 이 점에 대한 위반행위를 발견하면 즉시 보고하여야 한다.



CLIENT

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 SK hynix

CONTRACTOR	 SK ecoplant	PROJECT NAME	M15X P14 Project
COMMODITY	IW	JOB NO.	22330D
LINE NO.	80A-IW-ST-C-9528	SH'T NO. / REV.	01 / 01

[illegible][illegible]

경고 : 본 도면은 SKHYN의 기업비밀 자료로서, 어떠한 경우에도 무단 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사 내의 모든 직원, 계약자, 협력자, 대리인, 하위 계약자, 자문자, 기타 제3자에게도 이의 내용을 알리고, 이의 내용을 철저히 비밀로 관리할 것을 요구합니다. 이의 내용을 철저히 비밀로 관리하지 않으면, SKHYN은 법적 조치를 취할 수 있습니다.

PUNCH LIST

Project No : 22330D
Project Title : M15X Ph3 Project

Package No :
Date :

S/N	Drawing No	Description	Category A B C	Punch Clear Status			
				작성자		확인자	
				Name	Sign	Name	Sign
1							
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Remarks

A : Before Hydro Test
B : Before M/C
C : Before Commissioning